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Investigating the effect of welding tool length on mechanical strength of welded metallic matrix by molecular dynamics simulation

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A R T I C L E I N F O

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Mechanical behavior

A B S T R A C T

The welding process and the properties of welding instruments may improve the mechanical performance of an item. One of these properties is the length of the welding tool. This approach has a substantial effect on the mechanical strength of the metallic matrix. The current study used molecular dynamics modeling and LAMMPS software to evaluate the effect of welding tool length on the mechanical properties of a welded Cu–Ag metallic matrix. This simulation makes use of the Lennard-Jones potential function and the embedded atom model. First, the equilibrium phase of modeled samples was verified by changing the computation of kinetic and total energies. Next, the mechanical properties of the welded matrix were studied using the stated Young's modulus and ultimate strength. The stress-strain curve of samples demonstrated that the mechanical strength of atomic samples increased as the length of the welding tool (penetration depth) increased. Numerically, by increasing the tool penetration depth of Fe tools from 2 Å to 8 Å, Young's modulus and ultimate strength of the matrixes sample increase from 34.360 GPa to 1390.84 MPa to 38.44 GPa and 1510 MPa, respectively. This suggested that the length of the Fe welding tool significantly affected the mechanical properties of the welded metallic matrix. The longer the length of Fe welding tools, the more particles were involved, and consequently, more bonds were formed among the particles. Bonding among the particles caused changes in mechanical properties, such as greater ultimate strength. This method can optimize mechanical structures and be useful in various industries.

1. Introduction

Welding is one of the methods of connecting materials (metal, non-metal, composite, etc.). Two pieces are placed together and heated until they melt. In welding, in addition to heat, it also needs pressure in particular conditions. Heat can be generated using a gas flame, electric arc, friction, ultrasound waves, etc. [1]. There are two types of welding: fusion and solid. Fusion welding is a welding process (WP) that uses the melting of the base metal to create a weld [2], and there are different types of solid welding, including friction, friction stir, resistance, explosion, etc. [3]. Friction stir welding (FSW) is a solid-state technique that joins materials, particularly metals [4]. It generates frictional heat between a rotating tool and the workpiece, which softens the material and allows for plastic deformation [5]. The tool then moves along the joint, stirring the softened material and creating a solid-state bond [6]. FSW is known for its ability to produce high-quality, defect-free welds with excellent mechanical properties (MP).

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Abbreviations

MP	Mechanical properties
MM	Metallic matrix
CAMM	Cu–Ag metallic matrix
YM	Young's modulus
US	Ultimate strength
MD	Molecular dynamics
FSW	Friction stir welding
WP	Welding process

Regarding the high importance and benefits of welding, especially friction welding, many studies were conducted in these fields. Some researchers studied WP, and others studied the factors affecting the MP of metals in welding. In the meantime, some talked about penetration depth and its effect. Wu et al. [7] studied the effect of pulse TIG welding thermal cycle on the microstructure and MP of titanium/steel explosion welding. Bandhu et al. [8] evaluated the geometry of the weld bead in the metal arc WP for low-alloy steel with modified shortcircuit gas. Cheepu et al. [9] investigated the effect of joint interface on MP in dissimilar friction welds. The optimal hardness values were documented in the mixed zone and dynamic recrystallization zone. Anandaraj et al. [10] analyzed the effect of process parameters on the hot tensile behavior of rotary friction welding. Tang et al. [11] showed that transferring the surface friction regime from the coulomb to the shear friction regime is preferable for temperature homogenization in the welding interface. Elsheikh et al. [12] studied optimized artificial intelligence proposed to model the ultrasonic welding of a polymeric material blend. Then, the obtained experimental data were used to train the developed model. Kumar et al. [13] investigated the frictional properties of stir-welded polylactic acid/aluminum composites primed by fused filament fabrication. Sehgal et al. [14] investigated the effect of a variable welding current with constant voltage on the effect of resistance of the weld bead of similar and different materials. Agrawal et al. [15] studied the effect of WP parameters, such as voltage, current, welding velocity, and wire feeding were studied, and tests were conducted to check the strength and hardness of the connection. Zhang et al. [16] analyzed and briefly reviewed the challenges and solutions of advanced welding. The article discusses the difficulties in achieving optimal solutions and evaluates the current attempts made to improve them. Zou et al. [17] studied welding penetration depth control using relative fluctuation coefficient as feedback. They showed that an adaptive control algorithm could obtain uniform weld penetration depth. Cavilha et al. [18] investigated the effect of different power modulation formats (square, triangle, and trapezoid) on penetration while maintaining the same input energy (ratio of power to welding velocity). The results show that it is possible to achieve a 77 % improvement in penetration after power modulation for the energy of 120 kJ/m and square wave, which is more than what is found in the literature. Dos Santos Paes et al. [19] investigated the sensitivity analysis and multi-objective optimization of tungsten inert gas welding based on numerical simulation. Through the implementation of multi-objective optimization, a compromise was achieved between the penetration rate and HAZ volume.

Prior research showed that many factors, including voltage, current, power, and others, affect FSW. Molecular dynamics (MD) simulation is a computer method used to study atomic and molecular-level properties and actions of molecules and materials. MD simulation uses classical mechanics to anticipate the behavior of atoms and molecules by simulating their motion over time. Using MD simulation to study the effect of welding tool length on the mechanical strength of welded metallic matrix (MM) provides a way to study the process at a very small scale, allowing for a more detailed understanding of the underlying mechanisms. This can help researchers optimize the WP and develop new materials with improved properties. However, for the first time, the present research studied the effect of the length of welding tools on the MP of welded Cu–Ag metallic matrix (CAMM) by MD simulation. Fe welding tool length was 22, 25, 27, and 29 Å to check the changes in MP of CAMM after WP. The study analyzed the changes in the stress-strain curve, Young's modulus (YM), and the ultimate strength (US) of the CAMM.

2. Simulation and method

2.1. Simulation details

This paper investigated the effect of different welding tool lengths on the MP of CAMM. These samples were welded by a Fe welding tool with various lengths of 22, 25, 27, and 29 Å. The rotation period of the welding tool is set to 0.05 fs? Moreover, the linear velocity of the tool was 0.5 Å/fs. These settings caused the optimum welding process in our modeled atomic sample. Fe welding tool was modeled in a cylinder shape with Avogadro software. Fe tool modeled as BCC crystal structure with 2.87 Å as lattice constant. The simulation box was modeled with $110 \times 70 \times 50$ Å^3 dimensions and periodic boundary conditions. Using the NVT ensemble, the initial temperature was optimized to 300 K. The temperature damping value was 0.01 fs and the time step value was 0.1 fs, as shown in Table 1, the computational configuration of our MD simulations. Fig. 1 depicts a schematic of CAMM and Fe welding tools with different lengths, and Fig. 2 shows a schematic of CAMMs after the WP.

There were two phases to the study. Through kinetic and total energy changes, as well as NVT ensemble, CAMM achieved equilibrium in the first stage. Subsequently, welded CAMM was stretched in x direction using an external force (of 1 s^{-1} ratio) and simulated nanostructure's stress-strain curve, YM, and US were examined. A schematic of CAMM is shown in Fig. 3 both before and after the mechanical test. To report the mechanical performance of the simulated sample, the stress-strain curve output is important. Computationally, the stress value is calculated with $\frac{NkBT}{V} + \frac{1}{Vd}\sum_{i=1}^{N'} \overrightarrow{r}_i . \overrightarrow{f}_i$. Where N is the number of atoms in the system, V is the volume of the modeled system, m is atomic mass, r/v and f are the position/velocity and force vector of each atom. In this equation, N' defines atoms from neighboring subdomains. By this calculation, proportional strain and stress values are printed every 10000-time steps, and mechanical output is reported.

3. Results

In order to confirm equilibrium in CAMMs, the kinetic and total energy changes were examined in the next step. The kinetic and total energy changes across 100,000 time steps are shown in Figs. 4 and 5. The numerical findings indicate that total energy and kinetic energy converged to −8250.40 eV and 353.71 eV, respectively. Otherwise, these figures show a reduction in fluctuations across the simulation duration as the values converged to a constant value. Therefore, the

Table 1
MD simulation settings in current research.

Computational Parameter	Parameter Ratio/Setting
MD Box Size	$110 \times 70 \times 50$ Å^3
Number of Atoms	2799–2862
Boundary Condition	Periodic
Thermostat	Nose-Hoover
MD Ensembles	NVT
Initial Temp	300 K
Time Step	0.1 fs
Temperature Damping Ratio	0.01
Period of Welding Tool	0.05 fs
Linear Velocity of Welding Tool	0.5 Å/fs

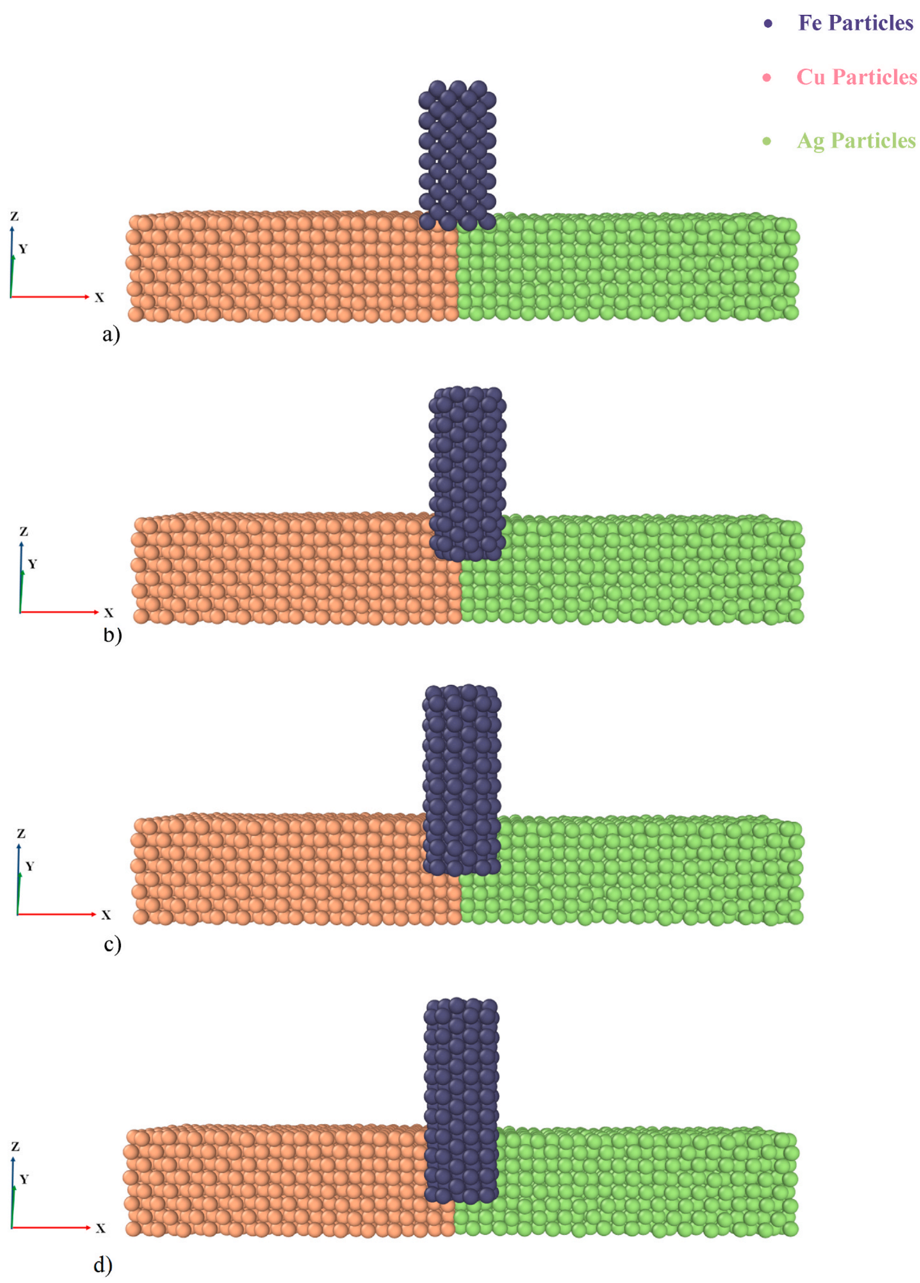

Fig. 1. A schematic of the CAMM with different welding tool lengths of a) 22, b) 25, c) 27, and d) 29 Å.

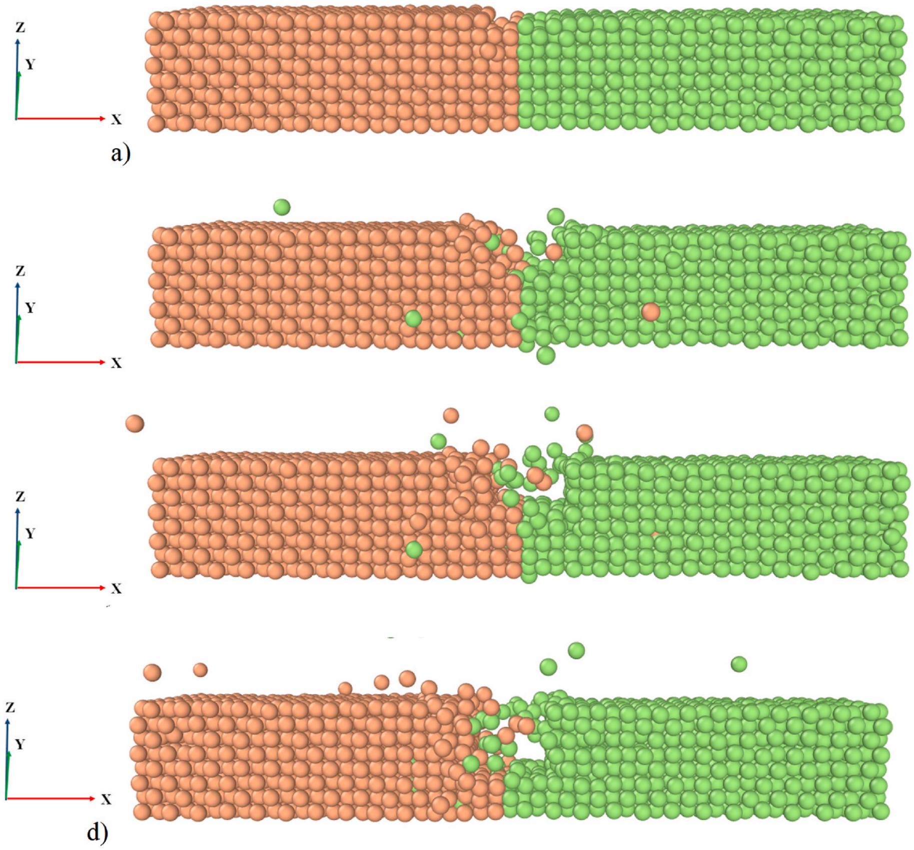

Fig. 2. A schematic of the CAMM after WP with different welding tool lengths of a) 22, b) 25, c) 27, and d) 29 Å.

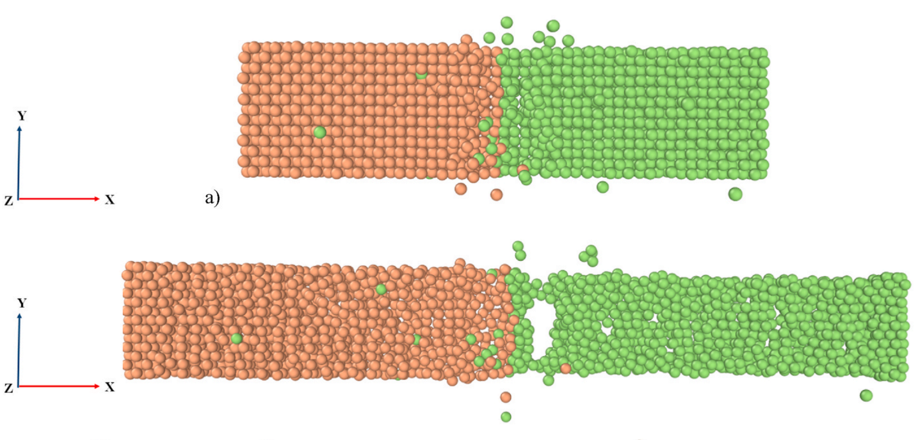

Fig. 3. A schematic of the CAMM before and b) after the mechanical test.

potential function was selected appropriately, and the simulation duration for the equilibration process was adequate.

It computed the radial distribution function (RDF) of Ag and Cu samples independently after identifying the equilibrium phase. RDF is a metric that quantifies the dispersion of particles in a substance around a reference particle. It measures the likelihood of discovering a particle at a certain separation from the reference particle. It plotted the MD outputs for this parameter in Fig. 6, and its results were consistent with previously reported structures for Ag and Cu samples [27–29]. This validates the present computational approach and provides confidence in the accuracy of the present results. Previous research studies utilized this approach, which further supports the validity of the present methodology.

4. Discussion

The sample was subjected to mechanical test settings after the identification of the equilibrium phase. Fig. 7 illustrates the progression of the mechanical testing procedure with a penetration depth of 2 Å. The findings show that YM was at 34.360 GPa and US was at 1390.84 MPa. It makes sense that the sample's MP would benefit from the welding tool's penetration depth being increased. As the penetration depth of the welding tool increases, more particles are involved, and more bonds are created among them. So, it can increase the strength of CAMM. It should be mentioned that it considers the simulation tool as rigid. If the tool particles stick in the welding region, the average distance among the particles in WP increases. Therefore, by not considering the welding

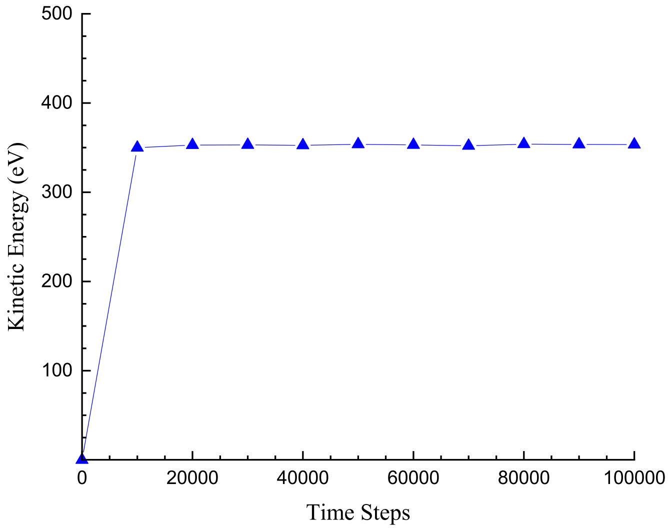

Fig. 4. The change of kinetic energy during 100,000-time steps.

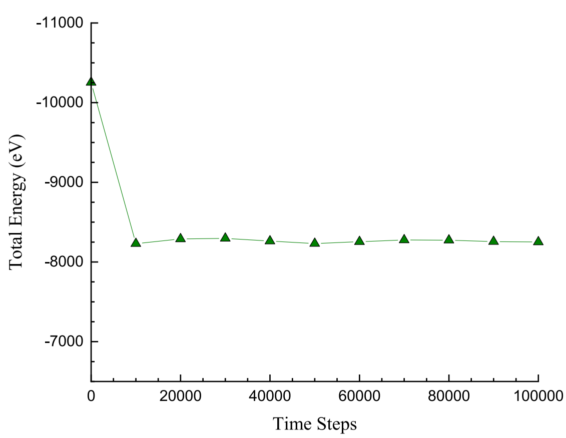

Fig. 5. The change of potential energy during 100,000-time steps.

tool's rigidity, the tool's atoms (iron) will penetrate the welding region, and this penetration will increase the atomic distance and decrease the attraction among atoms in the welding region. Consequently, the mechanical resistance of the material will decrease. To ensure the correctness of the simulated process in this research, comparing the MD outputs and previous experimental reports will be appropriate. The values obtained for YM for the welded sample (Ag–Cu system) consist of previous experimental reports for each pure sample [30,31]. It can be said that this comparison indicated the correctness of the simulation process carried out in the current research. Also, this described welded structure shows lesser YM than pure Cu and Ag. This behavior arose from lesser attraction energy/force between Cu and Ag atoms in the welding region. The interatomic energy in the welding region was calculated to analyze this description atomically. This calculation was done for the first time and described the estimated atomistic performance of Ag and Cu atoms in the welding region. Numerically, interatomic interaction in Ag–Ag, Cu–Cu, and Ag–Cu particles converged to −3216.0173, −3647.2691, and −174.74871 eV, respectively. These numerical outputs concluded that the WP decreased the interaction energy among welded samples. So, in actual cases, the mechanical performance of final structures converged to pure samples by improving interactions among welded samples.

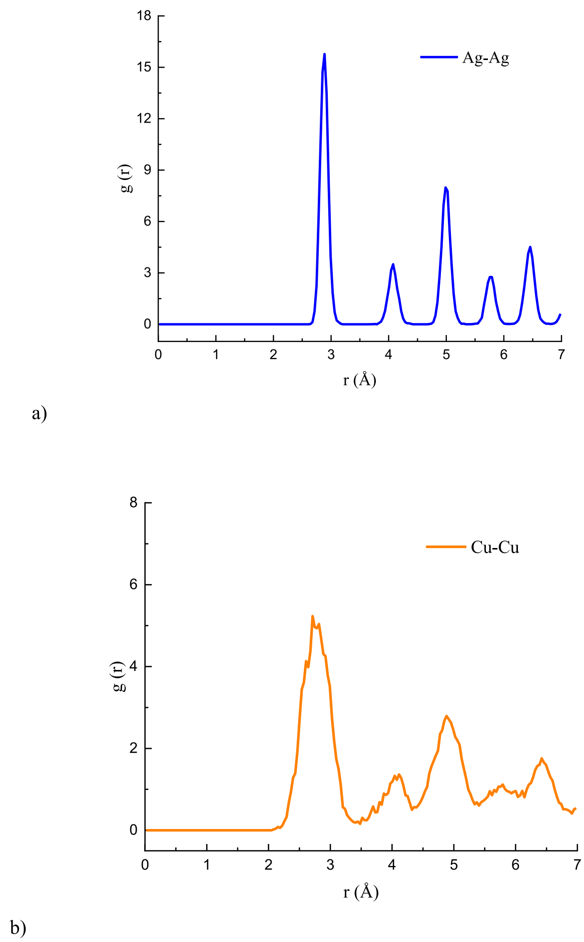

Fig. 6. The RDF output of a) Ag and b) Cu samples after equilibrium phase detection inside the computational box.

To complete the WP, the length of Fe welding tools was extended in the subsequent steps. Subsequently, the MP of CAMM was examined. The changes in the YM and stress-strain curve of CAMMs with penetration depth adjustments to 4, 6, and 8 Å are shown in Fig. 8. The findings show that the ideal ratio for these curves was at a depth of 8 Å. As mentioned before, as the penetration depth of the welding tool increased, more particles were involved, and more bonds were created among them. So, it can raise the strength of CAMM. Physically, the interaction energy calculation of modeled samples should clarify their mechanical performance. Numerically, the interaction energy of welded samples with 2, 4, 6, and 8 Å penetration depth converged to −7038.0351, −6668.8468, −6485.9027, and −6185.8522 eV after the WP. The convergence of this physical parameter in modeled samples showed the atomic changes did not diverge after the WP. This evolution arose from the appropriate locating of Ag and Cu atoms in the welding region. So, the designed atomic process in this computational research can be used in actual cases. The present calculated values for interaction energy among various samples before/after WP are reported in Table 2.

The change in YM and US of CAMMs after WP with different tool lengths are shown in Figs. 9 and 10 The optimized ratio of YM and US reach to 38.44 GPa and 1510 MPa, respectively, with a depth of 8 Å. The change in YM and the US of the CAMM at the different tool penetration depths are shown in Table 3. Hence, by increasing the depth of tool penetration, the mechanical strength of the sample increased, and MP became optimized. This solution can be helpful in various industries to access structures with the optimized mechanical performance of common structures after the WP.

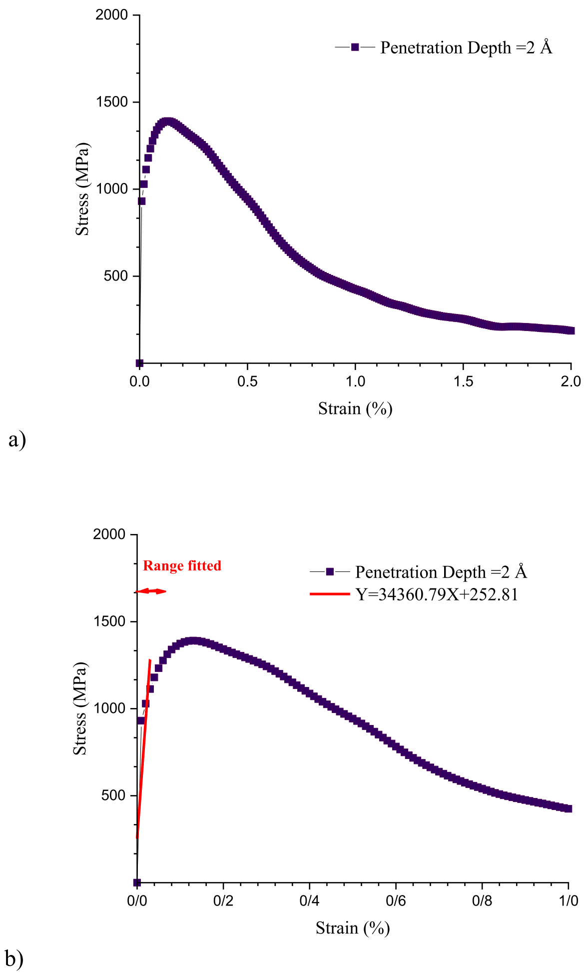

Fig. 7. The change in the a) stress-strain curve and b) YM of CAMMs.

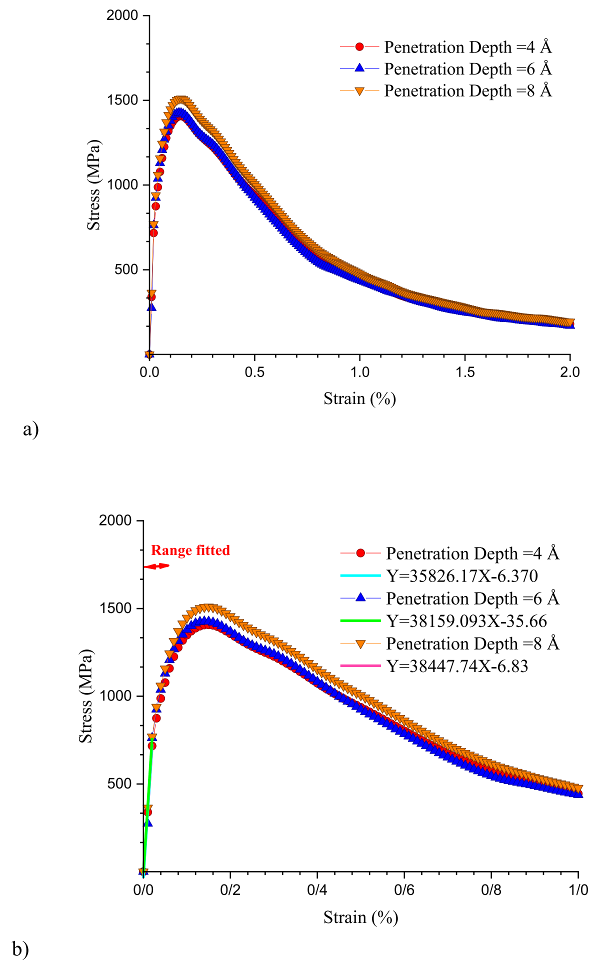

Fig. 8. The change in the a) stress-strain curve and b) YM of CAMMs.

5. Conclusion

Using MD simulation, this article examined the effect of welding tool length on the MP of a welded CAMM; this was an innovative method of research. By enabling the examination of the process at an extremely microscopic level, MD simulation provides a more comprehensive understanding of underlying mechanisms. In addition, this novel approach to examining the effect of welding instrument length was not the subject of extensive research. The main results of the current investigation can be categorized as follows.

- Simulated CAMMs' kinetic energy and total energy converge to 353.71 eV and −8250.40 eV after 1 ns (100,000-time steps), respectively.
- The ratio of YM and US of the welded CAMM (with welding Tools with 2 Å penetration depth) was calculated to be 34.360 GPa and 1390.84 MPa, respectively.
- By increasing depth penetration from 2 Å to 8 Å, YM and US of the welded matrix increase to 38.447 GPa and 1510 MPa, respectively.
- The penetration depth of 8 Å was the optimized ratio of YM and US. As the depth increased, particles increases, creating more continuity in the CAMM. Therefore, more cohesion increases strength.

In summary, atomistic samples effectively changed their mechanical performance with different penetration depths of welding tools, and a greater depth can effectively optimize mechanical performance. This research can potentially lead to new insights and optimizations in the WP and could contribute to developing new materials with improved properties.

Table 2
The interaction energy (between all atoms) changes as a function of penetration depth.

Penetration Depth (Å)	Before WP(eV)	After WP (eV)
2	−7097.1598	−7038.0351
4	−7097.1598	−6668.8468
6	−7097.1598	−6485.9027
8	−7097.1598	−6185.8522

CRediT authorship contribution statement

Xuejin Yang: Supervision, Writing – review & editing, Conceptualization, Data curation, Formal analysis. **Rassol Hamed Rasheed:** Supervision, Writing – review & editing, Conceptualization, Data curation, Formal analysis. **Sami Abdulhak Saleh:** Writing – review & editing, Conceptualization, Data curation, Formal analysis. **Mohammed Al-Bahrani:** Supervision, Writing – review & editing, Conceptualization, Data curation, Formal analysis. **C Manjunath:** Investigation, Writing – original draft. **Raman Kumar:** Investigation, Writing – original draft. **Soheil Salahshour:** Formal analysis, Writing – original draft. **Rozbeh Sabetvand:** Conceptualization, Data curation, Formal analysis.

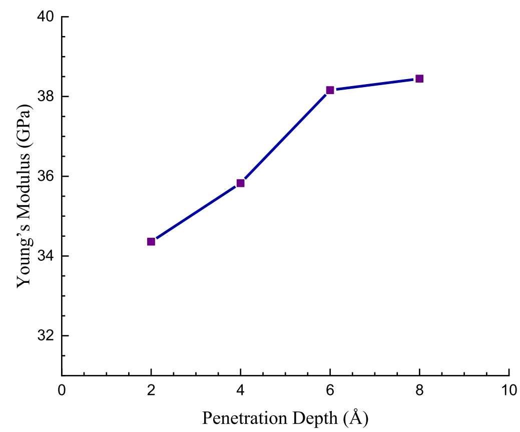

Fig. 9. The change in YM of CAMM with increasing the Fe tool length.

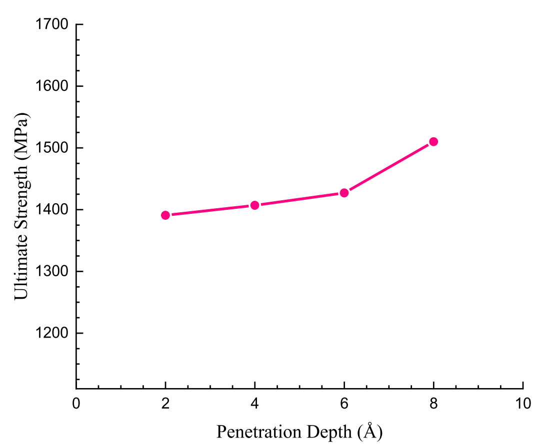

Fig. 10. The change in the US of the CAMM with increasing the Fe tool length.

Table 3
The change in the US and YM of simulated structure at the different penetration depths of the welding tool.

Penetration Depth (Å)	YM (GPa)	US (MPa)
2	34.360	1390.84
4	35.826	1407.04
6	38.159	1426.96
8	38.447	1510

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The authors listed on the manuscript are not employed by a government agency that has a primary function other than research and/or education. Also, the authors declare they are not belong to an official representative or on behalf of the government.

Data availability

No data was used for the research described in the article.

Appendix

A. MD Simulation

MD can investigate the thermodynamic behavior of materials using force, velocity, and position of particles. It can be briefly defined as step-by-step numerical solution of Newton's equation of motion for a many-particle system. In the Newtonian interpretation of the dynamics of motion, the translational motion of a spherical particle i arose from force F_i applied by an external agent [20].

$$F_i = m_i a_i = -\nabla_i U = -\frac{dU}{dr_i} \tag{a-1}$$

The most eminent factor in MD is the applied force on each particle. The applied force on a particle depends on its position relative to other particles in the MD method. The coupling property makes the analytical description of particles difficult. Therefore, integration equations of motion are used to solve this problem. This method divides the whole simulation time into small femtosecond intervals. Therefore, integration is done in equal and very small steps. The velocity-Verlet algorithm is a fast method for numerical integration at this stage [20,21].

$$r_i\,(t+\Delta t) = 2\,r_i(t) - r_i(t-\Delta t) + \left(\frac{d^2 r_i}{dt^2}\right)(\Delta t)^2 \tag{a-2}$$

$$v_i(t+\Delta t) = v_i(t) + \Delta t\, v_i(t) + \frac{\Delta t\,(a_i\,(t) + a_i(t+\Delta t))}{2} \tag{a-3}$$

In the next stage, MD uses the potential energy of a system. The potential function is divided into two potentials: interaction and non-interaction. The bonding and non-bonding interactions applied to the particles must be added together [20].

$$E_{total} = E_{bonded} + E_{nonbonded} \tag{a-4}$$

The non-bonded potential function occurred in terms of the particles in space. These functions had different forms, and in the following article, two of them were used: Lennard-Jones (LJ) and EAM potential functions. The LJ potential is as follows [22].

$$U_{LJ} = 4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right] \quad r_{ij} < r_c \tag{a-5}$$

here, r_c represents cut-off radius, σ_{ij} shows the distance where the potential became zero, r_{ij} indicates the distance of particles from each other, and ε_{ij} represents the depth of the potential. These are calculated as follows [23]:

$$\varepsilon_{ij} = \sqrt{\varepsilon_i \varepsilon_j} \tag{a-6}$$

$$\sigma_{ij} = \frac{\sigma_i + \sigma_j}{2} \tag{a-7}$$

The values of σ_{ij} and ε_{ij} coefficients for all present particles are reported in Table a-1.

Table a-1
The LJ potential function's parameters for the CAMM [24,25].

Particle's type	ε_{ij} (kcal/mol)	σ_{ij} (Å)	r_c(Å)
Ag	0.036	3.148	12
Cu	0.005	3.495	12
Fe	0.013	2.912	12

EAM potential was widely used to simulate metal particles and metal alloys. The simulations made by this potential were appropriate and consistent with the experimental results [26].

$$U_i = F_\alpha\left(\sum\nolimits_{i \neq j} \rho_\beta(r_{ij})\right) + \frac{1}{2}\sum_{i \neq j} \varphi_\beta(r_{ij}) \tag{a-8}$$

In the above equation, r_{ij} represents the distance of particles from each other, ρ_β shows attractive force, and φ_β represents repulsive force.

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